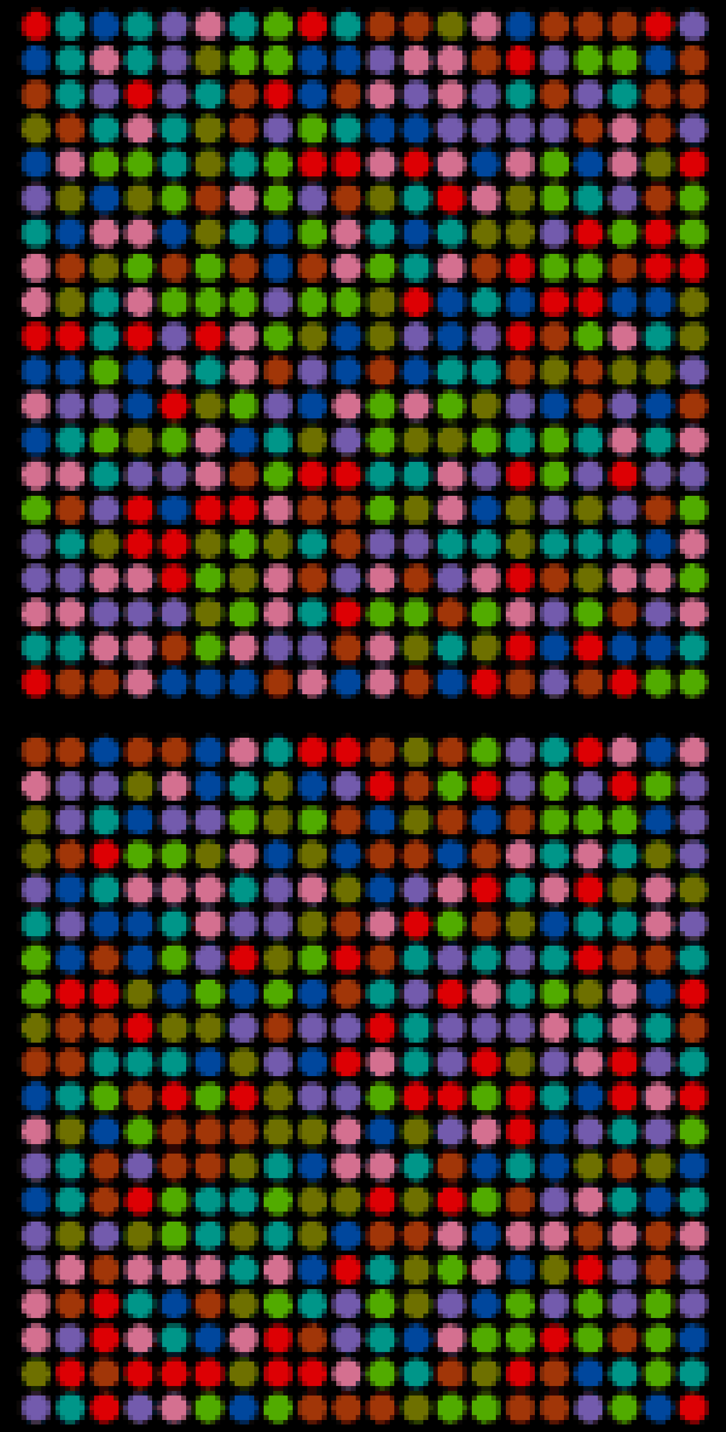
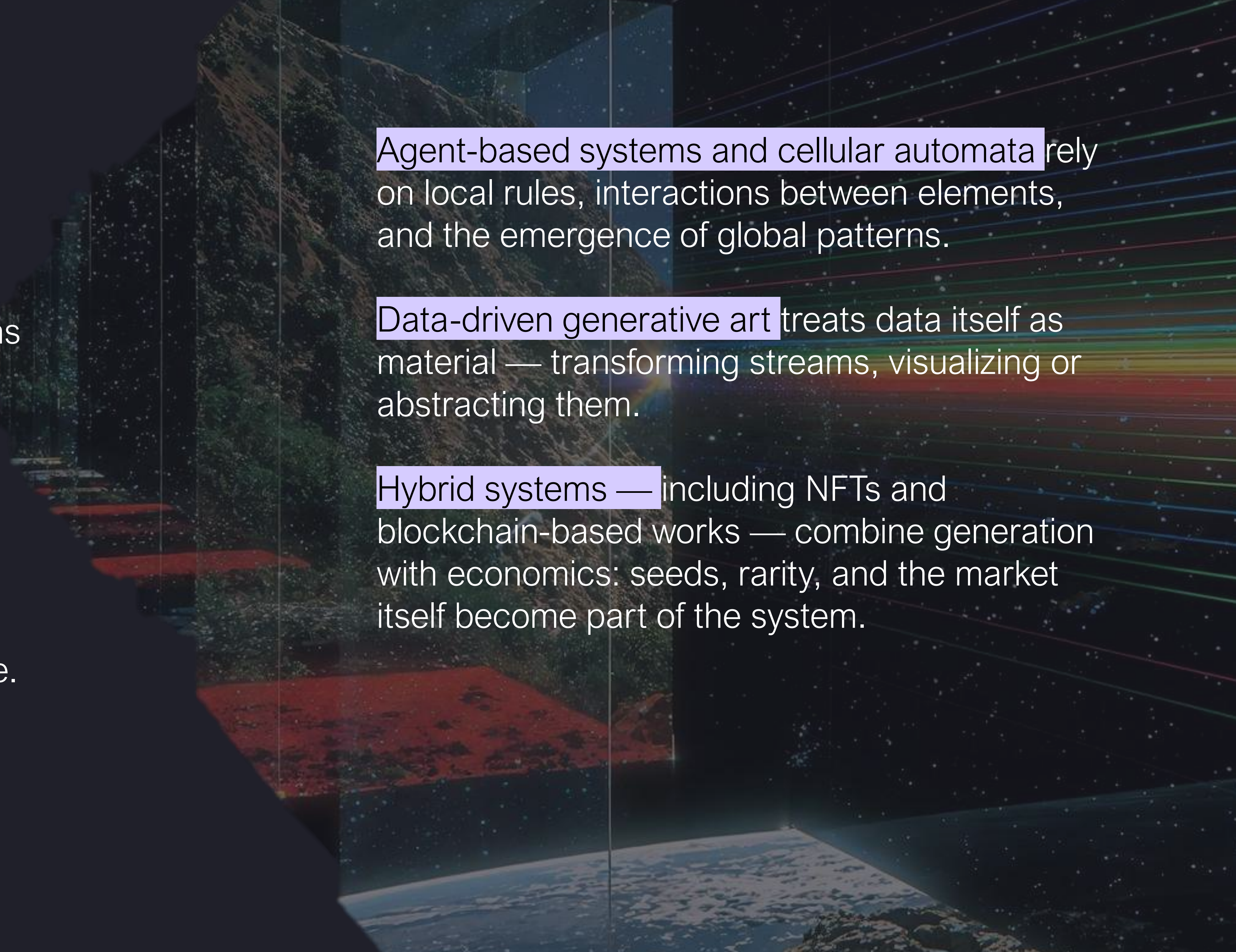
Aleatoric randomness is pure chance — like a dice throw — where a single random act directly determines the outcome.

Stochastic processes work with probability distributions. The system does not choose anything at all — it chooses within a defined range.

Pseudo-randomness is randomness generated by an algorithm rather than by true chaos.

In emergence randomness appears not because we introduced it, but because interactions within the system become opaque.





Procedural generative art operates through procedures, steps, instructions, and actions unfolding over time.

Algorithmic generative art works with formalized rules, repetition, transformation, and code — or mental code.

Agent-based systems and cellular automata rely on local rules, interactions between elements, and the emergence of global patterns.

Data-driven generative art treats data itself as material — transforming streams, visualizing or abstracting them.

Hybrid systems — including NFTs and blockchain-based works — combine generation with economics: seeds, rarity, and the market itself become part of the system.

Aspect	Cellular Automata (Internal Time)	Data-Driven (Archival Time)
Source of Form	Rules → Emergent Complexity	Data → Algorithm
Role of Time	Generator (tick → transformation)	Material ("scrolling" the archive)
Viewer	Observes the process	Immerses in the already-passed
Control	Artist sets rules	Artist curates data/prompt